Shashank C

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EXPERIENCE

Junior Associate - Operations — **November 2024 – November 2025**

Unifycx — ***Bangalore***

- Used **MobaXterm** to access **Linux** hosting servers and troubleshoot WordPress environment issues **theme/plugin conflicts** and apply **command-line fixes** to improve **site stability**. Performed **Linux log analysis** and **configuration checks** to diagnose and resolve server errors **403/404/500/internal server errors**, restoring **uptime efficiently**.
- Handled **DNS** troubleshooting and hosting configuration validation using **Linux**, resolving **FTP/DNS misconfigurations** and ensuring correct routing and connectivity. Administered **FTP/SFTP** user accounts and access controls to enable secure file transfers across **shared/dedicated hosting** environments.

PROJECTS

CI/CD GitOps 3-Tier Microservices Platform | github-link

- **Technologies: AWS EKS, Kubernetes, Jenkins, ArgoCD, Docker, Gateway API, Trivy, SonarQube, GitHub, Linux.**
- **Microservices Containerization**: Engineered optimized **multi-stage Dockerfiles** for **MERN** stack services to **minimize image size** and attack surface; orchestrated a **Docker Compose** environment to ensure environment parity between **local development** and **production**.**Managed Cluster Provisioning**: Utilized **eksctl** to provision a managed Amazon **EKS cluster**, facilitating the **rapid deployment** and **validation** of a complex **3-tier microservices architecture**.
- **Secure CI Pipelines:** Engineered automated **Jenkins pipelines** featuring **SonarQube code analysis** and **Trivy vulnerability scanning** to enforce a **security-first deployment model**.Integrated **GitHub Webhooks** for automated, event-driven **CI triggers**.**GitOps Continuous Delivery:** Deployed **ArgoCD** for **automated state synchronization** and drift detection from private repositories, maintaining a **3-minute polling interval** to ensure the cluster state precisely matches the Git-defined source of truth.
- **Security Secret Hygiene:** Secured the end-to-end workflow using **imagePullSecrets** for private registry authentication and **GitHub PATs** for encrypted access; enforced a strict **Zero-Hardcoding Policy** using Kubernetes **ConfigMaps** and **Secrets** to manage sensitive settings at runtime.**Traffic Management:** Deployed **Envoy API Gateway** using Kubernetes Gateway API resources (GatewayClass, HTTPRoute) to manage ingress traffic and provide a **centralized**, **secure entry point** for microservices.

Enterprise GitOps Platform: Automated EKS Infrastructure and 3-Tier Microservices | github-link

- **Technologies: AWS (VPC, EC2, IAM, EKS, EBS), Terraform, Linux, Kubernetes, Bash Scripting.**
- **Infrastructure as Code:** Transitioned from managed **CLI-based** provisioning (**eksctl**) to building a **Custom-Architected EKS Cluster** from scratch using **Terraform codebase**, **eliminating manual configuration** and ensuring **100 percent environment reproducibility** of complex infrastructure across **multiple AWS regions**.**Security and Isolation:** Implemented **secure** internet **ingress/egress** architecture: **Internet Gateway** for public subnets and **NAT Gateways** in the public subnets to provide secure outbound access for resources in the two Private subnets; enforced a strict **No Public IP policy** for **worker nodes** to **eliminate external attack vectors**.Engineered a **Bastion Host (Jump Server)** within the public subnet as the singular entry point for **administrative access**, enabling secure **SSH tunneling** to **EKS nodes** residing in private subnets.
- **Identity and Access Management (IAM):** Provisioned granular IAM roles and policies for the EKS Control Plane and worker nodes, adhering to the **Principle of Least Privilege(PoLP)**.**Persistent Storage and Data Durability:** Configured **AWS EBS CSI Driver** and **IAM OIDC providers** to enable **StatefulSets** to mount external **EBS volumes**; ensured **data persistence** by decoupling storage from the **ephemeral lifecycle of pods and nodes**, **preventing data loss** during **scaling** or **failures**.
- **Automated Scalability**: Implemented a dual-layer scaling strategy by deploying the **Kubernetes Metrics Server** and **Horizontal Pod Autoscaler (HPA)** for **application-level elasticity**. Integrated the **AWS Cluster Autoscaler** via **Helm** and **IAM OIDC roles** to dynamically **scale EC2 worker nodes** based on real-time resource demands, optimizing both **cluster performance** and **cloud expenditure**.**Advanced Deployment Strategies**: Implemented a **Canary Deployment model** to orchestrate **risk-reduced application updates**; utilized **traffic-splitting** to route a subset of users to the **new version**, ensuring **zero downtime** and enabling **seamless rollbacks** while maintaining **high application availability**.

Enterprise EKS Monitoring System | github-link

- **Technologies: Prometheus, Grafana, Alertmanager, Node Exporter, Blackbox Exporter, AWS, Linux.**
- **Out-of-Cluster Observability:** Architected an independent monitoring stack on a secure **Bastion host** for **high availability**. Configured **cross-subnet network rules** to **securely scrape metrics** from **private EKS nodes**.**Infrastructure Telemetry:** Deployed **Node Exporter** as a Kubernetes **DaemonSet** across all worker nodes; configured **Prometheus** to perform **service discovery** and scraping of hardware metrics across the private network.
- **Proactive Alerting Logic:** Developed custom **PromQL alerting rules** for infrastructure health and resource exhaustion, utilizing time-series sensitivity to **eliminate false positives** and ensure **actionable incident reporting**.**Synthetic Endpoint Monitoring:**Integrated **Blackbox Exporter** to perform continuous **HTTP probing** of both the **Jenkins CI/CD** dashboard and the public-facing **application endpoint (via Envoy API Gateway)**, ensuring **real-time detection** of **service unavailability** for both internal and end-user traffic.
- **Incident Response Pipeline:** Configured Alertmanager with **SMTP** integration to route **critical-severity alerts** to **engineering teams**, ensuring **rapid incident acknowledgment** and resolution.**Visual Analytics:** Operationalized **Grafana dashboards** to visualize **cluster health**, providing an intuitive interface for **capacity planning** and **real-time performance monitoring**.

TECHNICAL SKILLS

- **linux, Github, Bash, Python, Docker, Jenkins, ArgoCD, Ansible, AWS, kubernetes, Terraform, Prometheus, Grafana.**

EDUCATION

Master of computer application — **Graduated: 2024**

Pes university, Bangalore — ***CGPA: 7.5 / 10***